PW VIDYAPEETH

IIT-JEE | NEET | FOUNDATION

12th
JEE Main

ROI

Milestone Test-01

Phase-02

DURATION: 180 Minutes | **DATE: 08/06/2025** | **M.MARKS: 300**

Topic Covered

Physics : Electric Charges and Fields

Chemistry : Solutions

Mathematics : Determinants

GENERAL INSTRUCTION

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours** duration.
3. The test booklet consists of 75 questions. The maximum marks are **300**.
4. There are three Sections in the question paper, Section I, II & III consisting of Section-I **(Physics)**, Section-II **(Chemistry)**, Section-III **(Mathematics)** and having **25 questions** in each part in which first **20** questions are of Objective Type and Last **5 questions** are integers type and all **25 questions are compulsory**.
5. There is only one correct response among 4 alternate choices provided for each objective type question.
6. Each correct answer will give **4** marks while **1** Mark will be deducted for a wrong response.
7. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
8. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

OMR Instructions

1. Use blue/black dark ballpoint pens.
2. Darken the bubble completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be ready by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple marking will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**
8. In integer answer type if the answer is 590 then fill it in OMR as 5 in Hundreds place (H row), 9 in Tens place (T row) and 0 in Ones place (O row).

EXAMPLE:
H ⓪①②③④●⑥⑦⑧⑨
T ⓪①②③④⑤⑥⑦⑧●
O ●①②③④⑤⑥⑦⑧⑨

Name of the Student (In CAPITALS) : ____________________

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Roll Number : ____________________

OMR Bar Code Number : ____________________

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Candidate's Signature **Invigilator's Signature**

SECTION-I (PHYSICS)

Single Correct Type Questions (1-20)

1. Which configuration will have zero net electric field at the center of a triangle?
(1) Equal charges at all three corners
(2) $+Q$ at one corner, $-Q$ at other two
(3) $-Q$, $-Q$ and zero charge
(4) $+Q$, $-Q$ and zero charge

2. An electric dipole is placed at an angle of 30° with an electric field of intensity 10^4 NC^{-1}. It experiences a torque equal to 8 N-m. Calculate the magnitude of charge on the dipole, if the dipole length is 4 cm.
(1) 8×10^3 C
(2) 8×10^{-2} C
(3) 4×10^{-2} C
(4) 4×10^{-3} C

3. A charge of 6 μC is to be divided into two parts. The distance between the two divided charge is constant. The magnitude of the divided charges so that the force between them is maximum, will be:
(1) 1μC and 5μC (2) 3μC and 3μC
(3) 2μC and 4μC (4) 2.5μC and 3.5 μC

4. For a uniformly charged ring of radius R = 2m, the electric field on its axis has the largest magnitude at a distance h from its centre. Then value of h is
(1) $2\sqrt{2}$ m (2) $\frac{2}{\sqrt{3}}$ m
(3) $\sqrt{2}$ m (4) $\frac{1}{\sqrt{2}}$ m

5. An electric dipole consists of two opposite charges, each of magnitude 2.0 μC separated by a distance of 4.0 cm. The dipole is placed in an external electric field of 10^5 NC^{-1}. The maximum torque on the dipole is $n\times10^{-3}$ Nm. Find n.
(1) 1
(2) 8
(3) 3
(4) 4

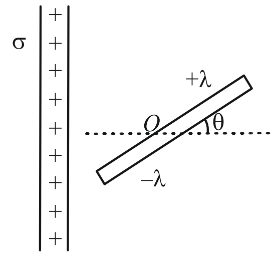
6. The net electric field at the center of a regular hexagon due to six equal charges placed at the vertices is:
(1) Zero
(2) Directed toward one vertex
(3) Along a side
(4) Infinity

7. Two unit negative charges are placed on straight line. A positive charge q is placed exactly at the mid-point between these unit charges. If the system of these three charges is in equilibrium, the value of q (in C) is x. Find $4x$
(1) 1 (2) 2
(3) 3 (4) 4

8. An electron initially at rest falls a distance of 1.5 cm in a uniform electric field of magnitude 2×10^4 N/C. The time taken by electron to fall this distance is $x\times10^{-9}$ sec. Find x.
(1) 2.92 (2) 3.42
(3) 4.42 (4) 4

9. Two identical charges repel each other with a force equal to 40 N when they are 0.6 m apart in air. The value of each charge is (in μC)
(1) 40 (2) 30
(3) 20 (4) 60

10. A large sheet carries uniform surface charge density σ. A rod of length 2ℓ has a linear charge density λ on one half and – λ on the second half. The rod is hinged at mid-point O and makes an angle θ = 30° with the normal to the sheet. The torque experienced by the rod is



(1) $\frac{\sigma\lambda\ell^2}{6\varepsilon_0}$ (2) $\frac{\sigma\lambda\ell^2}{4\varepsilon_0}$
(3) $\frac{\sigma\lambda\ell^2}{2\varepsilon_0}$ (4) $\frac{\sigma\lambda\ell^2}{8\varepsilon_0}$

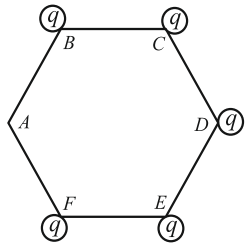
11. Electric field on axis of a charged ring radius R, charge Q, at distance x from center of ring is
(1) $\frac{Q}{4\pi\varepsilon_0\left(x^2+R^2\right)}$
(2) $\frac{Qx}{4\pi\varepsilon_0\left(x^2+R^2\right)^{3/2}}$
(3) $\frac{Q}{4\pi\varepsilon_0 x^2}$
(4) zero

12. A given charge situated at a certain distance from a short electric dipole on axial position, experiences a force F. If the distance of charge is half, in the same direction the force acting on the charge will be

(1) $\frac{F}{27}$ (2) $\frac{F}{8}$

(3) $\frac{F}{64}$ (4) $\frac{F}{9}$

13. As shown five identical charges are kept at five vertices of a regular hexagon. Match the following two columns at centre of the hexagon. If in the given situation electric field at centre is $5\frac{\text{N}}{\text{C}}$. Then,



Column I		Column II	
A	If charge at B is removed, then electric field will become	P	$10\frac{\text{N}}{\text{C}}$
B	If charge at C is removed, then electric field will become	Q	$5\frac{\text{N}}{\text{C}}$
C	If charge at D is removed then electric field will become	R	$0\frac{\text{N}}{\text{C}}$
D	If charges at B and C both are removed, then electric field will become	S	$5\sqrt{3}\frac{\text{N}}{\text{C}}$

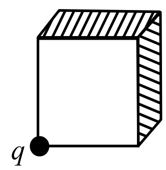
	A	B	C	D
(1)	Q	S	P	R
(2)	S	Q	R	P
(3)	P	R	Q	S
(4)	R	P	S	Q

14. Total electric flux coming out of a unit positive charge put in air is $\frac{x}{\varepsilon_0}$ (in SI unit). The value of x is

(1) 2 (2) 3

(3) 4 (4) 1

15. The length of each side of a cubical closed surface is ℓ. If charge q is situated on one of the vertices of the cube. The flux passing through shaded face of the cube is.



(1) $\frac{q}{24\varepsilon_0}$ (2) $\frac{q}{6\varepsilon_0}$

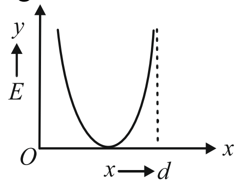
(3) $\frac{q}{8\varepsilon_0}$ (4) $\frac{q}{12\varepsilon_0}$

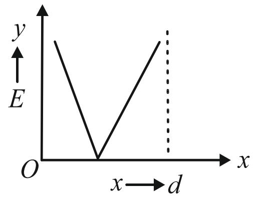
16. Two opposite and equal charges, each of magnitude 2×10^{-8} coulomb when placed 2×10^{-2} cm away, form a dipole. If this dipole is placed in an external electric field 4×10^{8} N/C, the value of maximum torque is $x\times10^{-4}$ Nm. Find x.

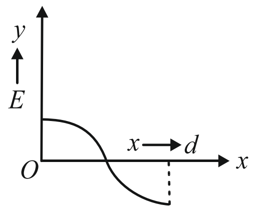
(1) 16 (2) 32

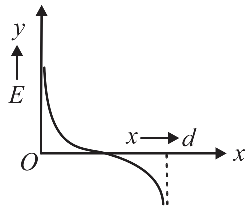
(3) 48 (4) 64

17. Two identical charges are placed at a separation of d, one is at origin and other at distance d along x-axis. P is a point on the line joining the charges, at a distance x from any one charge. The field at P is E, E is plotted against x for values of x from close to zero to slightly less than d. Which of the following best represents the resulting curve

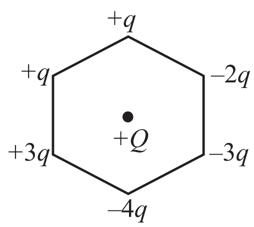
(1) 

(2) 

(3) 

(4) 

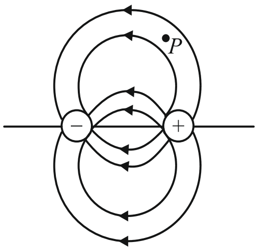
18. Six charges are kept at the vertices of a regular hexagon as shown in the figure. If magnitude of force applied by $+Q$ on $+q$ charge is F, then magnitude of net electric force on the $+Q$ is



(1) $4F$ (2) $9F$

(3) $5F$ (4) $6F$

19. Figure shows the electric field lines around an electric dipole. Which of the arrows best represents the electric field at point P?



(1) ↘ (2) ↖

(3) ↗ (4) ↙

20. Four charges $+Q$ at square corners side a, force on one charge is

(1) zero

(2) $\frac{Q^2}{4\pi\varepsilon_0 a^2}\sqrt{3}$

(3) $\frac{Q^2}{4\pi\varepsilon_0 a^2}\left(\sqrt{2}+1\right)$

(4) $\frac{Q^2}{4\pi\varepsilon_0 a^2}\left(\sqrt{2}+\frac{1}{2}\right)$

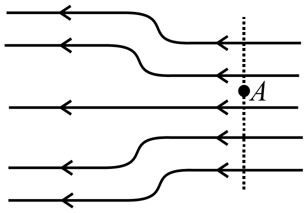
Integer Type Questions (21-25)

21. The magnitude of electric field at a distance r from an infinitely long thin rod having a linear charge density λ is $\frac{x\lambda}{2\pi\varepsilon_0 r}$. Find x

22. Two point charges $+9e$ and $+81e$ are at 16 cm away from each other. Where should another charge q be placed between them so that the system remains in equilibrium. The distance will be x cm from $+9e$. Find x

23. Three charges each of $+1\mu C$ are placed at the corners of an equilateral triangle. If the force between any two charges be 10 N, then the net force on either charge will be $\sqrt{x}$ N. Find x

24. In the electric field shown in figure, the electric lines in the left have twice the separation as that between those on right. If the magnitude of the field at point A is 20 NC^{-1}, the magnitude of force experienced by an electron placed at point A is $x \times 10^{-19}$ N. Find x.



25. A charge of 1 μ C is divided into two parts such that their charges are in the ratio of 1:2. These two charges are kept at a distance 1m apart in vacuum. Then, the electric force between them is $n \times 10^{-3}$ N. Find n.

SECTION-II (CHEMISTRY)

Single Correct Type Questions (26-45)

26. Among the following mixtures, dipole-dipole as the major interaction is present in

(1) Benzene and ethanol

(2) KCl and water

(3) Acetonitrile and acetone

(4) Benzene and CCl_4

27. 100 cc of 0.3 M H_2SO_4 and 200 cc of 0.3 M HCl were mixed together. The normality of the mixed solution will be

(1) 0.2 N

(2) 0.4 N

(3) 0.8 N

(4) 0.6 N

28. Which of the following is not correct for ideal solution?

(1) $\Delta V_{mix} = 0$

(2) $\Delta H_{mix} = 0$

(3) $\Delta S_{mix} = 0$

(4) Obeys Raoult's law

29. When W_A g solute (molecular mass M_A) dissolves in W_B g solvent (molecular mass M_B), the molality M of the solution is

(1) $\frac{W_B}{M_B} \times \frac{1000}{W_A}$ (2) $\frac{W_A}{M_A} \times \frac{1000}{W_B}$

(3) $\frac{W_B}{M_A} \times \frac{M_A}{1000}$ (4) $\frac{W_A}{W_B} \times \frac{M_B}{1000}$

30. The amount of anhydrous Na_2CO_3 present in 500 mL of 0.75 M solution is

(1) 3.975 g (2) 39.75 g
(3) 397.5 g (4) 3975 g

31. At high altitude, the boiling of water occurs at low temp. because

(1) Atmospheric pressure is low
(2) Temperature is low
(3) Atmospheric pressure is high
(4) Does not depend upon atmospheric pressure

32. The vapour pressure of two liquids P and Q are 90 torr and 70 torr respectively. The total vapour pressure obtained by mixing 2 mole of P and 3 mole of Q would be

(1) 78 torr (2) 20 torr
(3) 140 torr (4) 72 torr

33. A solution of two liquids boils at a temperature more than the boiling point of either of them. Hence, the binary solution shows

(1) Negative deviation from Raoult's law
(2) Positive deviation from Raoult's law
(3) No deviation from Raoult's law
(4) Positive or negative deviation from Raoult's law depending upon the composition

34. For an ideal binary liquid solution with $P_A^0 > P_B^0$. Which relation between X_A (mole fraction of A in liquid phase) and Y_A (mole fraction of A in vapour phase) is correct; X_B and Y_B are mole fraction of B in liquid and vapour phase respectively :

(1) $X_A = Y_A$
(2) $X_A > Y_A$
(3) $\frac{X_A}{X_B} < \frac{Y_A}{Y_B}$
(4) X_A, Y_A, X_B and Y_B cannot be correlated

35. On mixing 20 mL of acetone with 50 mL of chloroform, the total volume of the solution will be

(1) < 70 mL (2) > 70 mL
(3) = 70 mL (4) Cannot be predicted

36. A mixture of volatile components A and B has vapour pressure (in torr)
$P = 235 - 124\ X_A$
Where X_A is the mole fraction of A in mixture. Hence P_A^o and P_B^o are (in torr)

(1) 254, 119
(2) 111, 235
(3) 135, 254
(4) 124, 235

37. Azeotropic mixture are

(1) Constant boiling point mixture without changing the composition
(2) Those which boil at different temperatures
(3) Those having different composition in their vapour phase.
(4) Those which can be easily fractionally distilled.

38. **Statement 1:** Cooking time in pressure cooker is reduced
Statement 2: The boiling point of water decreases inside the pressure cooker

(1) Statement 1 is True, Statement 2 is True; Statement 2 is correct explanation for Statement 1
(2) Statement 1 is True, Statement 2 is True; Statement 2 is not correct explanation for Statement 1
(3) Statement 1 is True, Statement 2 is False
(4) Statement 1 is false, Statement 2 is True

39. The amount of ice that will separate out on cooling a solution containing 50 g of ethylene glycol in 200 g water to $-18.6°C$ will be (Molar mass of ethylene glycol = 62 and k_f of water = 1.86 °C kg mol^{-1})

(1) 80.64 g (2) 161.3 g
(3) 119.35 g (4) 38.7 g

40. If for an aqueous KCl solution elevation in boiling point is 0.2°C then what will be the elevation in boiling point in °C of aqueous $K_4[Fe(CN)_6]$ solution for the same molal concentration?
(Consider 100% dissociation of electrolytes)

(1) 0.1 (2) 0.2
(3) 0.6 (4) 0.5

41. **Assertion:** When raisins are placed in salt water, they will shrink.
Reason: It happens due to the phenomenon of osmosis.

(1) Both assertion and reason are correct, and the reason is the correct explanation for the assertion
(2) Both assertion and reason are correct, but the reason is not the correct explanation for the assertion
(3) The assertion is incorrect, but the reason is correct
(4) Both are assertion and reason are incorrect

42. Vapour pressure of CCl_4 at 25 °C is 142 mm of Hg and 0.4 g of a non-volatile solute (mol. wt = 65) is dissolved in 100 mL CCl_4. Find the vapour pressure of the solution. (Density of CCl_4=1.6 g/cm^3)
(1) 94.39 mm Hg (2) 141.93 mm Hg
(3) 141.16 mm Hg (4) 199.34 mm Hg

43. Solute A is a binary electrolyte and solute B is ternary-electrolyte. If 0.1 M solution of solute A produces an osmotic pressure of 2P, then 0.05 M solution of B at the same temperature will produce an osmotic pressure equal to: (consider 100% dissociation of electrolytes)
(1) P (2) 1.5 P
(3) 2 P (4) 3 P

44. Which is not correct about Henry's law?
(1) The gas in contact with the liquid should behave as an ideal gas
(2) There should not be any dissociation or association of the gas
(3) The pressure applied should not be too high
(4) The gas should react with the liquid (solvent)

45. Match the following:

Column-I		Column-II	
A	Hypertonic	P	Solutions having same osmotic pressure
B	Isotonic	Q	A solution having higher osmotic pressure than some other solutions
C	Iso-osmotic	R	Not related to osmotic pressure
D	Hypotonic	S	A solution having lower osmotic pressure than some other solution

	A	B	C	D
(1)	Q	P	P	S
(2)	R	S	Q	P
(3)	Q	R	S	Q
(4)	S	P	Q	R

Integer Type Questions (46-50)

46. The molality of a solution having 18 g of glucose (mol. Wt. = 180) dissolved in 500g of water is m. Find the value of 10 m.

47. How many grams of glucose must be present in 0.5 litre of a solution for its osmotic pressure to be same as that of 8 g glucose in 1 litre?

48. If 0.3 g of a solute (non-electrolyte), dissolved in 30 g of solvent, is boiled at a temperature higher by 0.108°C than that of the pure solvent. The molecular weight of the substance (in g/mol) (molal elevation constant for the solvent is 2.16k kg mol^{-1}) is

49. Aqueous solution of 0.004 M K_2SO_4 and 0.01 M urea are isotonic. The percentage degree of dissociation of K_2SO_4 is:

50. A solution containing 2.4 grams of sulphur atom in 100 g of a solvent (k_f = 8.0 K kg mol^{-1}) gave a freezing point depression of T°C, the molecular formula of sulphur in the solution is S_6. Find the value of T. (Assuming, 100% association of sulphur atoms.)

SECTION-III (MATHEMATICS)

Single Correct Type Questions (51-70)

51. The sum of the solutions of the equation $\begin{vmatrix} x & 2 & -1 \\ 2 & 5 & x \\ -1 & 2 & x \end{vmatrix} = 0$ is
(1) 2
(2) 4
(3) 6
(4) 8

52. If $f(\alpha) = \begin{vmatrix} 1 & \alpha & \alpha^2 \\ \alpha & \alpha^2 & 1 \\ \alpha^2 & 1 & \alpha \end{vmatrix}$, then $f\left(\sqrt[3]{2}\right)$ is equal to
(1) 1
(2) –1
(3) 4
(4) 2

53. If $\alpha, \beta, \gamma \in R$, then the determinant
$$\Delta = \begin{vmatrix} \left(e^{\alpha}+e^{-\alpha}\right)^2 & \left(e^{\alpha}-e^{-\alpha}\right)^2 & 4^2\alpha\beta\gamma \\ \left(e^{\beta}+e^{-\beta}\right)^2 & \left(e^{\beta}-e^{-\beta}\right)^2 & 4^2\alpha\beta\gamma \\ \left(e^{\gamma}+e^{-\gamma}\right)^2 & \left(e^{\gamma}-e^{-\gamma}\right)^2 & 4^2\alpha\beta\gamma \end{vmatrix}$$ is
(1) Independent of α, β and γ
(2) Dependent on α, β and γ
(3) Dependent on only two out of α, β, γ
(4) Dependent on only one out of α, β, γ

54. If $\Delta_1 = \begin{vmatrix} 1 & 1 & 1 \\ 2a & 2b & 2c \\ a^2 & b^2 & c^2 \end{vmatrix}$, $\Delta_2 = \begin{vmatrix} 1 & bc & 2a \\ 1 & ca & 2b \\ 1 & ab & 2c \end{vmatrix}$, then
(1) $\Delta_1 + \Delta_2 = 0$ (2) $\Delta_1 + 2\Delta_2 = 0$
(3) $\Delta_1 = \Delta_2$ (4) $\Delta_1 = 2\Delta_2$

55. If $\begin{vmatrix} -2a & a+b & a+c \\ b+a & -2b & b+c \\ c+a & b+c & -2c \end{vmatrix} = 2\alpha(a+b)(b+c)(c+a) \neq 0$, then α is equal to

(1) $a + b + c$ (2) abc
(3) 2 (4) 1

56. The value of determinant $\frac{1}{2}\begin{vmatrix} 0 & xy^2 & xz^2 \\ x^2y & 0 & yz^2 \\ x^2z & zy^2 & 0 \end{vmatrix}$ is

(1) $2xyz$ (2) $x^3y^3z^3$
(3) 0 (4) $4xyz$

57. If $f(x) = \begin{vmatrix} 2\left(x-\frac{1}{2}\right) & x-1 & x^2 \\ 1 & 0 & x \\ x & 1 & 0 \end{vmatrix}$, then the value of $f(2)$ is equal to

(1) 0 (2) 1
(3) 2 (4) 3

58. If the value of $\Delta = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 1+\sin\theta & 1 \\ 1+\cos\theta & 1 & 1 \end{vmatrix}$ is equal to P (where θ is real number) and $Q = -\sin\theta\cos\theta$ then $P + Q$ is equals to

(1) 1 (2) 2
(3) 0 (4) $\frac{1}{2}$

59. The value of the determinant $\begin{vmatrix} x & x+y & x+2y \\ x+2y & x & x+y \\ x+y & x+2y & x \end{vmatrix}$ is

(1) $9x^2(x+y)$
(2) $9y^2(x+y)$
(3) $-3y^2(x+y)$
(4) $-9x^2(x+y)$

60. If $\begin{vmatrix} y+z & x & y \\ z+x & z & x \\ x+y & y & z \end{vmatrix} = k(x+y+z)(x-z)^2$, then k is equal to

(1) $2xyz$
(2) 1
(3) xyz
(4) $x^2y^2z^2$

61. If one root of determinant $\begin{vmatrix} x & 3 & 7 \\ 2 & x & 2 \\ 7 & 6 & x \end{vmatrix} = 0$, is -9, then the other two roots are

(1) 2, 7 (2) 16, –7
(3) –16, 7 (4) –2, –7

62. If $a \neq p, b \neq q, c \neq r$ and $\begin{vmatrix} p & b & c \\ a & q & c \\ a & b & r \end{vmatrix} = 0$, then the value of $\frac{p}{p-a} + \frac{q}{q-b} + \frac{r}{r-c}$ is

(1) 2 (2) 3
(3) 4 (4) 5

63. If x, y, z are different from zero and $\begin{vmatrix} a & b-y & c-z \\ a-x & b & c-z \\ a-x & b-y & c \end{vmatrix} = 0$, then the value of the expression $\frac{a}{x} + \frac{b}{y} + \frac{c}{z}$ is

(1) 3 (2) 2
(3) 4 (4) 8

64. The value of $\begin{vmatrix} \log_5 729 & \log_3 5 \\ \log_5 27 & \log_9 25 \end{vmatrix} \begin{vmatrix} \log_3 5 & \log_{27} 5 \\ \log_5 9 & \log_5 9 \end{vmatrix}$ is equal to

(1) 1 (2) 6
(3) $\log_5 9$ (4) 4

65. If $\Delta_k = \begin{vmatrix} k & 1 & 5 \\ k^2 & 2n+1 & 2n+1 \\ k^3 & 3n^2 & 3n+1 \end{vmatrix}$, then $\frac{1}{2}\sum_{k=1}^{n} \Delta_k$ is equal to

(1) $2\sum_{k=1}^{n} k$ (2) $\sum_{k=1}^{n} k^2$
(3) $\frac{1}{2}\sum_{k=1}^{n} k^2$ (4) 0

66. If $x \neq 0$, $\begin{vmatrix} x+1 & 2x+1 & 3x+1 \\ 2x & 4x+3 & 6x+3 \\ 4x+4 & 6x+4 & 8x+4 \end{vmatrix} = 0$, then $\frac{x+1}{2}$ is equal to

(1) x (2) 0
(3) $2x$ (4) $3x$

67. If $\alpha+\beta=-\frac{b}{a}$, $\alpha\beta=\frac{c}{a}$ and let $\alpha^n+\beta^n=S_n$ for $n\geq 1$. Then, the value of the determinant $\begin{vmatrix} 3 & 1+S_1 & 1+S_2 \\ 1+S_1 & 1+S_2 & 1+S_3 \\ 1+S_2 & 1+S_3 & 1+S_4 \end{vmatrix}$ is

(1) $\frac{b^2-4ac}{a^4}$

(2) $\frac{(a+b+c)(b^2+4ac)}{a^4}$

(3) $\frac{(a+b+c)(b^2-4ac)}{a^4}$

(4) $\frac{(a+b+c)^2(b^2-4ac)}{a^4}$

68. If $\frac{a_2}{a_1}=\frac{a_3}{a_2}=\frac{a_4}{a_3}\text{...........}=\frac{a_n}{a_{n-1}}$, then the determinant

$\Delta=\begin{vmatrix} \log a_n & \log a_{n+1} & \log a_{n+2} \\ \log a_{n+3} & \log a_{n+4} & \log a_{n+5} \\ \log a_{n+6} & \log a_{n+7} & \log a_{n+8} \end{vmatrix}$ is equal to

(1) 2 (2) 4

(3) 0 (4) 1

69. The value of the determinant $\begin{vmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{vmatrix}$ is equal to λ then $(\sqrt{2})^{\log_2(\lambda)^2}$ is equal to

(1) 8 (2) 2

(3) 16 (4) 4

70. If $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}=5$, then the value of

$\begin{vmatrix} b_2c_3-b_3c_2 & c_2a_3-c_3a_2 & a_2b_3-a_3b_2 \\ b_3c_1-b_1c_3 & c_3a_1-c_1a_3 & a_3b_1-a_1b_3 \\ b_1c_2-b_2c_1 & c_1a_2-c_2a_1 & a_1b_2-a_2b_1 \end{vmatrix}$ is

(1) 25 (2) 36

(3) 20 (4) 15

Integer Type Questions (71-75)

71. Sum of real roots of the equation

$\begin{vmatrix} 1 & 4 & 20 \\ 1 & -2 & 5 \\ 1 & 2x & 5x^2 \end{vmatrix}=0$, is

72. The value of $\begin{vmatrix} 5^2 & 5^3 & 5^4 \\ 5^3 & 5^4 & 5^5 \\ 5^4 & 5^5 & 5^6 \end{vmatrix}$, is

73. Value of determinant $\begin{vmatrix} \log e & \log e^2 & \log e^3 \\ \log e^2 & \log e^3 & \log e^4 \\ \log e^3 & \log e^4 & \log e^5 \end{vmatrix}$ is equal to

74. If $\begin{vmatrix} x^n & x^{n+2} & x^{n+3} \\ y^n & y^{n+2} & y^{n+3} \\ z^n & z^{n+2} & z^{n+3} \end{vmatrix}=(y-z)(z-x)(x-y)\left(\frac{1}{x}+\frac{1}{y}+\frac{1}{z}\right)$

then the absolute value of n is equal to

75. If $\begin{vmatrix} a+b & b+c & c+a \\ b+c & c+a & a+b \\ c+a & a+b & b+c \end{vmatrix}=k\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$, then $\frac{k}{2}$ is equal to

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ROUGH WORK